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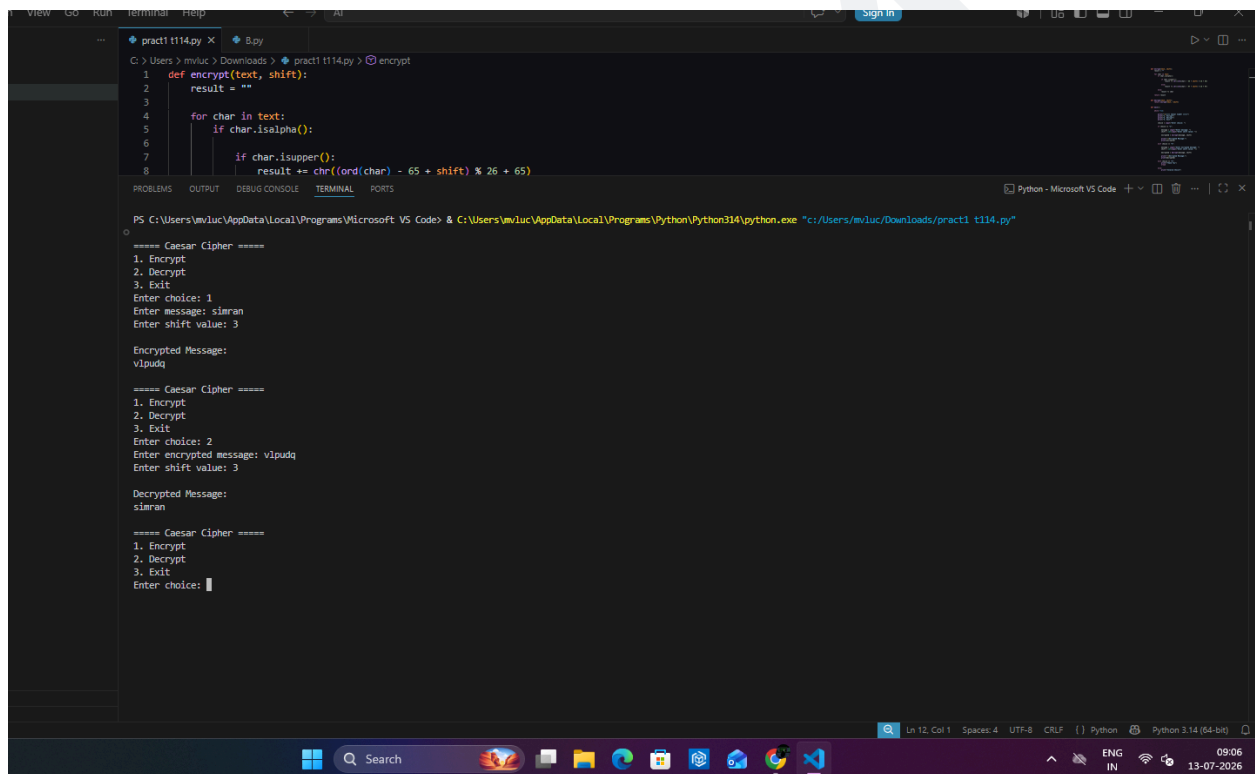
Practical no.1 Cyber security

1.A

Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

1. B

Design and implement algorithms to encrypt and decrypt messages using transposition techniques



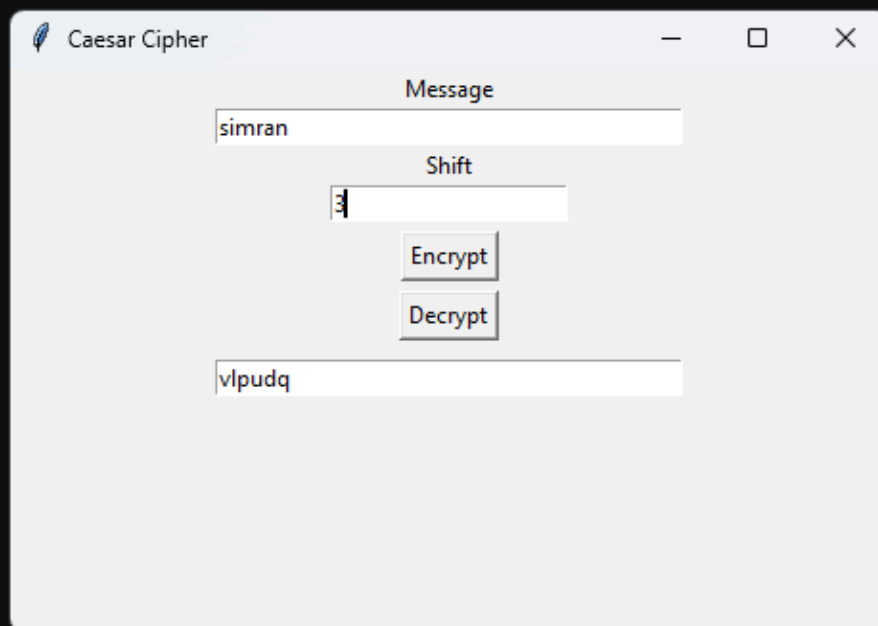
The image shows a screenshot of a Windows operating system with a code editor (VS Code) and a terminal window. The code editor displays a Python script named 'pract1 t114.py' which implements a Caesar Cipher algorithm. The script defines an 'encrypt' function that takes a string 'text' and an integer 'shift' as input. It iterates through each character in 'text', checks if it is an alphabetic character using 'char.isalpha()', and if so, shifts its ASCII value by 'shift' (wrapping around the alphabet). The 'decrypt' function is also defined, which shifts the encrypted message back by the same 'shift' value. The terminal window shows the execution of the script, where the user enters '1' to encrypt, 'siran' as the message, and '3' as the shift value. The output shows the encrypted message 'vlpudq'. The user then enters '2' to decrypt, enters the encrypted message 'vlpudq', and enters '3' as the shift value. The output shows the decrypted message 'siran'. The user then enters '3' to exit the program.

```
def encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            if char.isupper():
                result += chr((ord(char) - 65 + shift) % 26 + 65)
            else:
                result += chr((ord(char) - 97 + shift) % 26 + 97)
    return result

def decrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            if char.isupper():
                result += chr((ord(char) - 65 - shift) % 26 + 65)
            else:
                result += chr((ord(char) - 97 - shift) % 26 + 97)
    return result

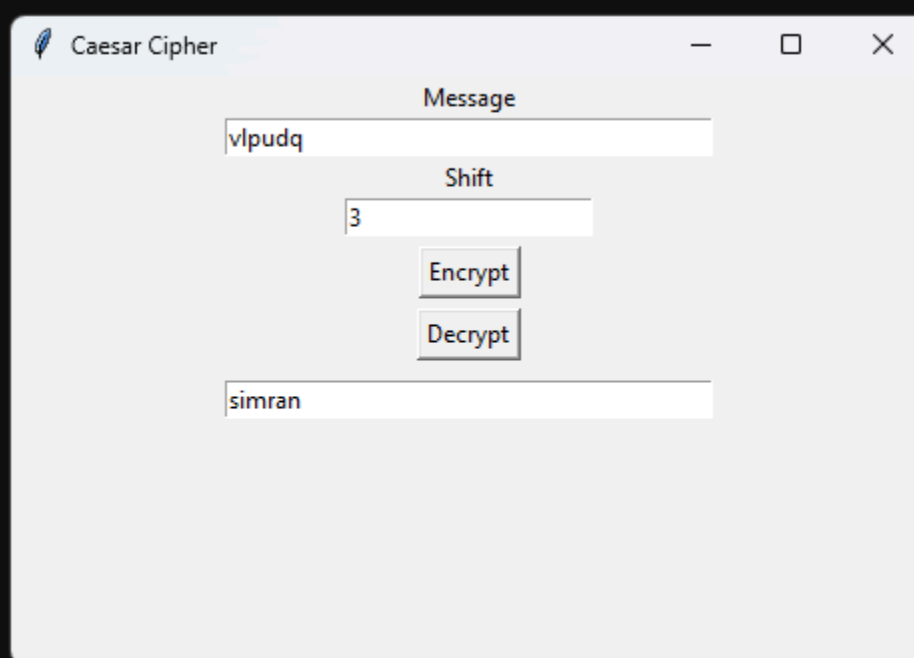
if __name__ == '__main__':
    while True:
        print("===== Caesar Cipher =====")
        print("1. Encrypt")
        print("2. Decrypt")
        print("3. Exit")
        choice = input("Enter choice: ")
        if choice == '1':
            message = input("Enter message: ")
            shift = input("Enter shift value: ")
            encrypted_message = encrypt(message, int(shift))
            print("Encrypted Message: ", encrypted_message)
        elif choice == '2':
            encrypted_message = input("Enter encrypted message: ")
            shift = input("Enter shift value: ")
            decrypted_message = decrypt(encrypted_message, int(shift))
            print("Decrypted Message: ", decrypted_message)
        elif choice == '3':
            break
```

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A screenshot of a Java Swing window titled "Caesar Cipher". The window has a light gray background and standard window controls (minimize, maximize, close) in the top right corner. It contains the following elements:

- A text field labeled "Message" containing the text "simran".
- A text field labeled "Shift" containing the number "3".
- Two buttons: "Encrypt" and "Decrypt", stacked vertically.
- A text field at the bottom containing the encrypted text "vlpudq".

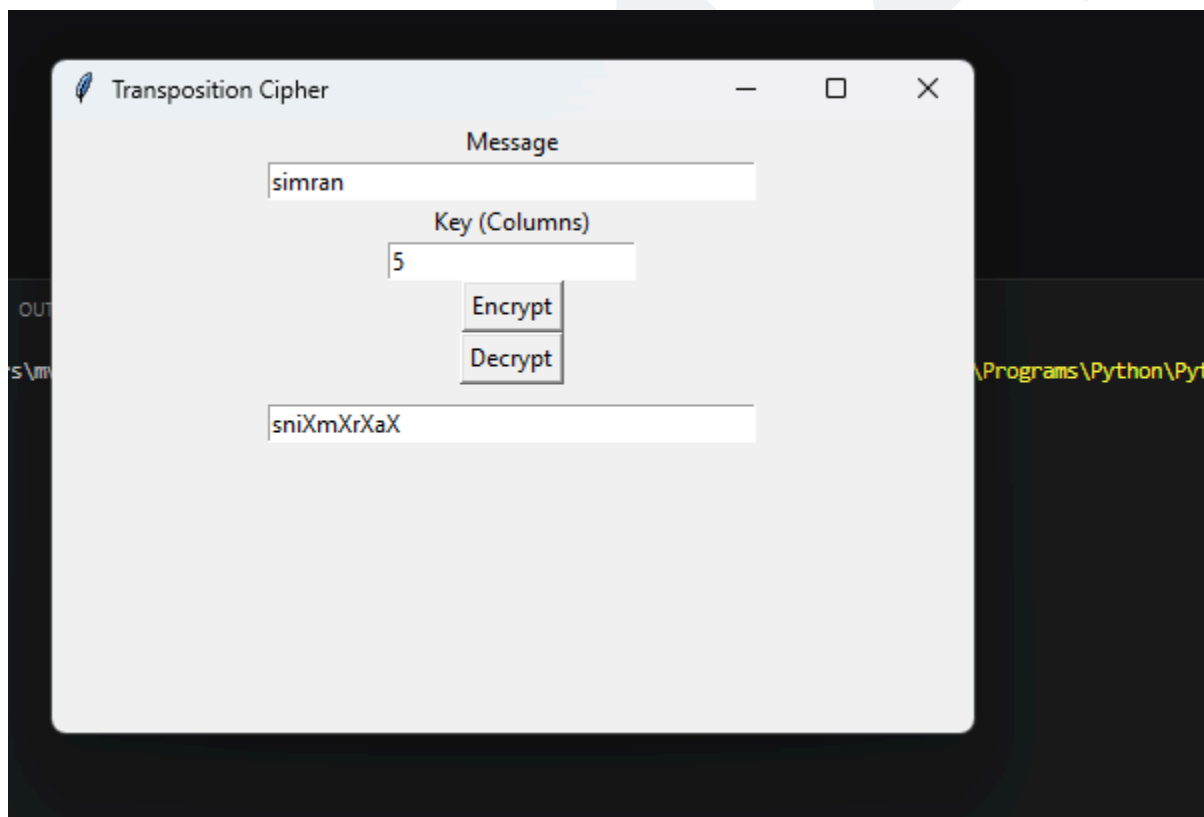


A screenshot of the same "Caesar Cipher" application window, but with the "Decrypt" button clicked. The state is as follows:

- The "Message" text field now contains the decrypted text "simran".
- The "Shift" text field still contains the number "3".
- The "Encrypt" and "Decrypt" buttons are still present.
- The bottom text field is now empty.

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```
7 message = message.replace(' ', '')
8 columns = key
9 rows = math.ceil(len(message) / columns)
10
11 matrix = []
12 index = 0
13
14 for i in range(rows):
15     for j in range(columns):
16         matrix.append(message[index])
17         index += 1
18
19 # Encrypting the message
20 encrypted_message = ''
21 for i in range(0, len(matrix), columns):
22     encrypted_message += matrix[i:i+columns]
23     encrypted_message += '\n'
24
25 # Decrypting the message
26 decrypted_message = ''
27 for i in range(0, len(encrypted_message), columns):
28     decrypted_message += matrix[i:i+columns]
29     decrypted_message += '\n'
30
31 # Main menu
32 while True:
33     print("\n===== Transposition Cipher =====")
34     print("1. Encrypt")
35     print("2. Decrypt")
36     print("3. Exit")
37     choice = input("Enter choice: ")
38     if choice == '1':
39         message = input("Enter Message: ")
40         key = input("Enter Key (Columns): ")
41         encrypted_message = encrypt(message, key)
42         print("Encrypted:\n", encrypted_message)
43     elif choice == '2':
44         cipher_text = input("Enter Cipher Text: ")
45         key = input("Enter Key (Columns): ")
46         decrypted_message = decrypt(cipher_text, key)
47         print("Decrypted:\n", decrypted_message)
48     elif choice == '3':
49         break
50     else:
51         print("Invalid choice. Please try again.")
```



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